

# MVC Task Business Requirements

## 1. Membership Business Rules

### 1.1. Create

1. A membership can only be created if the member exists in the system.
2. A membership can only be created if the plan exists in the system.
3. A member cannot have more than one Active membership at the same time.
4. Only active plans can be assigned
5. When a membership is created, its EndDate is automatically calculated based on the plan duration.
6. Membership status is computed: "Active" if EndDate > Now, else "Expired"

### 1.2. Delete

1. Cancellation Delete Memberships For Member On This Plan
2. A membership can only be deleted if it is Active.

## 2. Booking Business Rules

### 2.1. Create

1. A booking can only be created for a future session. Any session that has already started or finished cannot be booked.
2. Any action such as booking, cancellation, or marking attendance is not allowed if the referenced booking or session does not exist.
3. A member must have an Active membership in order to book a session.
4. A session must have available capacity. Booking is rejected if capacity is full.

5. When a booking is created, IsAttended is always set to false by default.
6. Member cannot book the same session twice

## 2.2. Member Attended

1. Attendance can only be marked for ongoing sessions (start date has passed but end date has not).

## 2.3. Cancel

1. Any action such as booking, cancellation, or marking attendance is not allowed if the referenced booking or session does not exist.
2. A booking can only be cancelled for future sessions. Once the session has started, cancellation is not allowed.